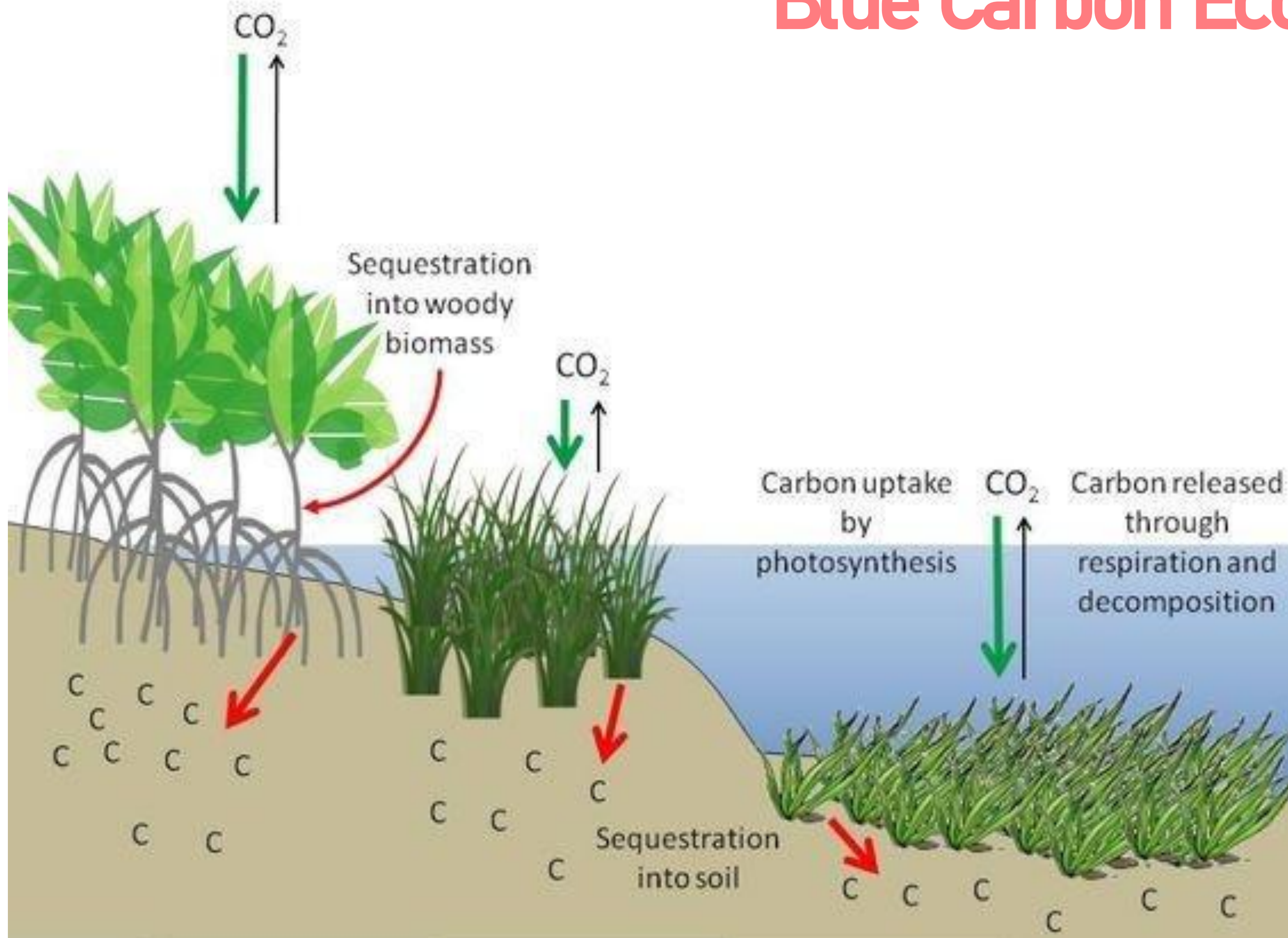


Lithoclimatic factors and microbial processes involved in soil carbon sequestration in mangrove ecosystems of Brazil

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Suresh et Park, « Blue carbon stock of Mangroves »

Blue Carbon Ecosystems

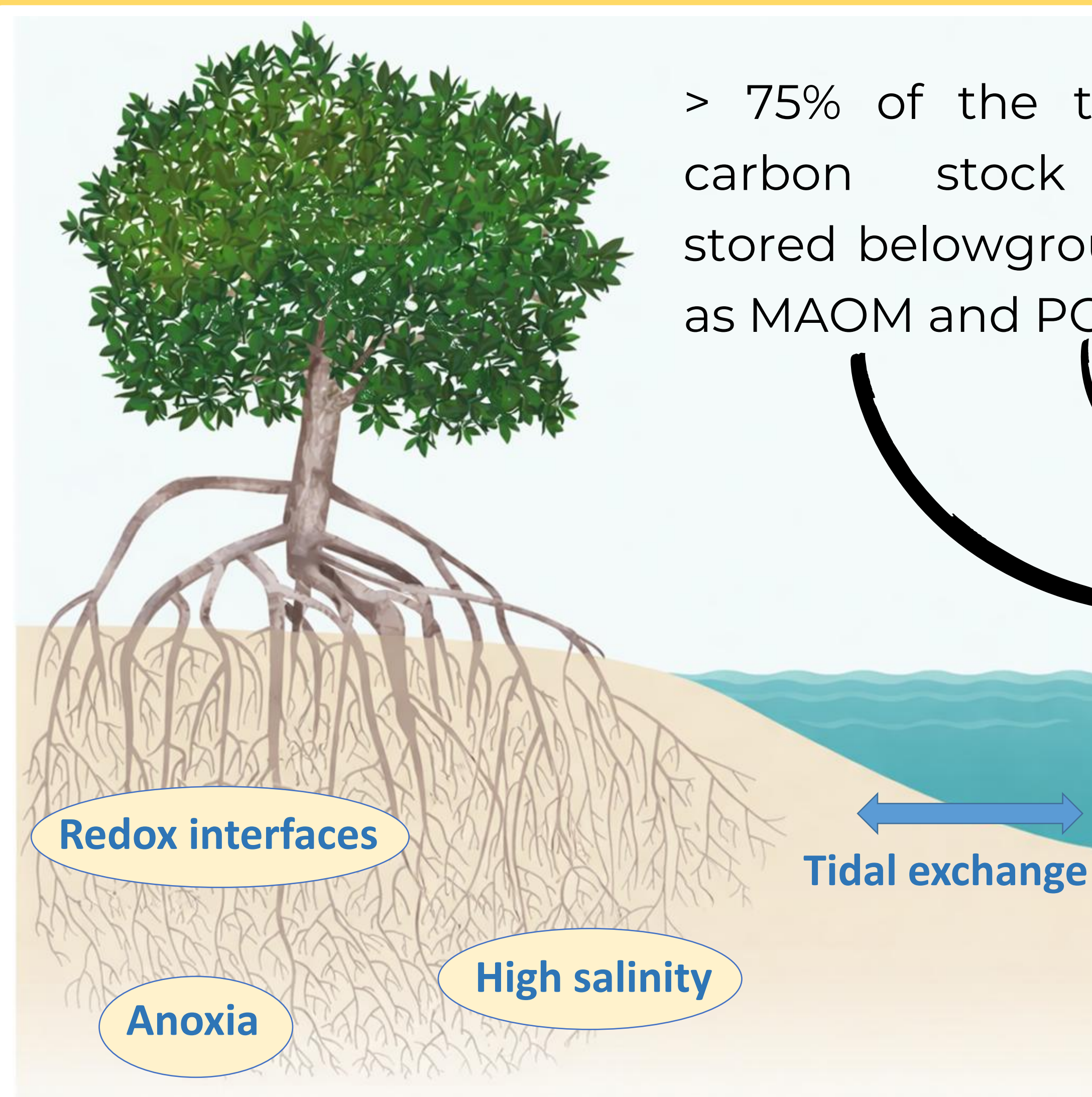


Mangrove **Salt marsh** **Seagrass meadow**

Carbon storage capacity occurs under harsh environmental conditions:

➔ high salinity ➔ anoxia ➔ tidal flooding

- Blue carbon ecosystems possess a unique capacity to sequester large amounts of carbon from both autochthonous and allochthonous sources
 - Can sequester carbon 10 times faster than terrestrial ecosystems
- However, they are highly threatened and suffer high losses every year:
 - urban and industrial development
 - drainage, pollution
 - pressures from agriculture and aquaculture
 - sea-level and temperature rise
- Yet, mechanisms controlling carbon stabilization in their soils remain poorly understood
- Gaining a deeper understanding of how carbon is captured and stored will enable the development of more effective management practices and policy guidelines to protect these vital ecosystems



> 75% of the total carbon stock is stored belowground as MAOM and POM

POM = Particulate Organic Matter

- Lightweight aboveground and belowground plant fragments, with a size > 53 μm and a density > 1,6 g/cm^3
 - Easily decomposable by microbes
 - Fast turnover (years to decades)
- ➔ **Dynamic carbon pool**

MAOM = Mineral-Associated Organic Matter

- Organic compounds bound to mineral surfaces, with a size < 53 μm and a density < 1,6 g/cm^3
 - Lots of MAOM complexes (Fe, Mn, Al, Clay)
 - Slower turnover (centuries up to millenia)
- ➔ **Physically and chemically protected carbon pool**

Scientific question

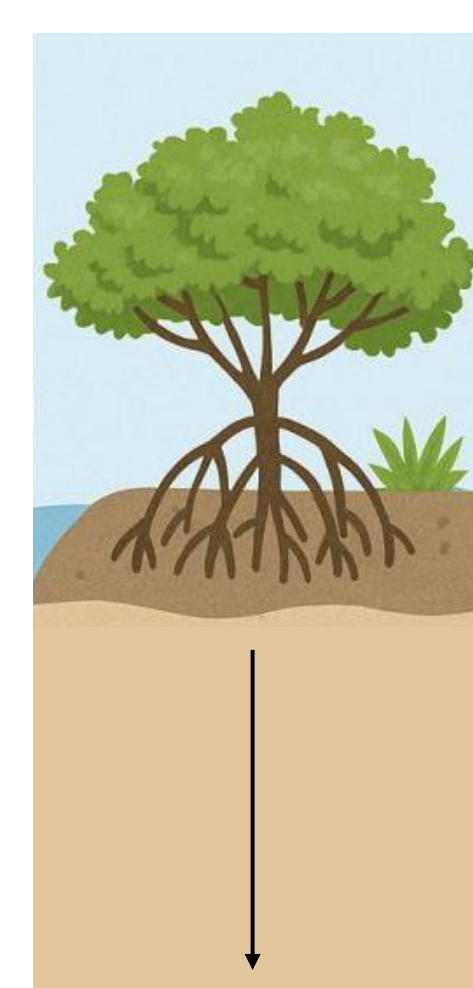
How do organic matter recalcitrance, redox conditions, microbial dynamics and soil geomorphology interact to control carbon stabilization in mangroves ?

Objectives

- Identify the mechanisms driving carbon stabilization
- Identify the proportion of MAOM and the different minerals present in the soils
- Assess how microbial composition and activity affect OM persistence
- Compare contrasted geomorphological settings

How?

By collecting samples:



➔ Along a salinity gradient

➔ Deep cores (> 1m)

➔ Across different climatic and geomorphological settings

